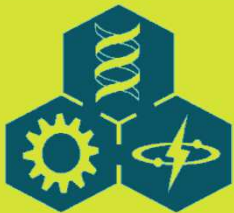
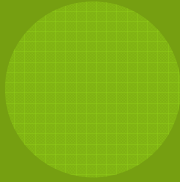


LECTURE 9:

MODEL ANALYSIS





OUTLINE

9.1 Analyzing Model Responses

9.2 Uncertainty Analysis

9.3 Analysis of Model Behavior

9.4 Mathematical Details



9.1 ANALYZING MODEL RESPONSES

- ◎ Validation is model analysis concerned with evaluating model quality relative to the real world using comparisons with empirical data.
- ◎ This chapter focuses on analyzing model performance by actively manipulating various components of the model.
- ◎ We will discuss two types of manipulations of the model structure.
 - ◎ The first type manipulates the equations and parameters of the model to ascertain the extent and effect of modeler uncertainty.
 - ◎ The second manipulation alters the values of state variables to determine if the system will return to the premanipulation levels.



9.2 UNCERTAINTY ANALYSIS

- ◎ Sources of Uncertainty in Biological Modeling
 - ◎ *Biological hypotheses and mathematical formulation.* We may be ignorant of the correct biological processes involved.
 - ◎ *Parameter values.* We may be ignorant of the mean and variance of the population from which our parameter estimates are drawn.
 - ◎ *Natural variation.* The system may have components that must be treated as stochastic (e.g., temperature). We will, therefore, be able to make only probabilistic predictions.



9.2 UNCERTAINTY ANALYSIS

- ◎ Sources of Uncertainty in Biological Modeling
 - ◎ There is little we can do about the uncertainty of biological hypotheses and mathematical formulation other than to learn more, design better experiments, and be more clever in our mathematical formulation.
 - ◎ The effect on our predictions of our uncertainty in parameter values can be investigated using a combination of *parameter sensitivity analysis* and *error analysis*.
 - ◎ For natural variation, we use *stochastic models* (models with output influenced by random effects on model variables or parameters) to address the problem.



9.2 UNCERTAINTY ANALYSIS

- ◎ Parameter Sensitivity Analysis
 - ◎ Parameter sensitivity analysis involves analyzing differences in model response to small differences in parameter values.
 - ◎ Author's interpretation: Parameter sensitivity analysis addresses the question of "What are the dynamical effects of modeler uncertainty about the true mean value of the parameters?"
 - ◎ Strictly speaking, we can ask this question of several model components: parameters, initial conditions, or driving variables.



9.2 UNCERTAINTY ANALYSIS

◎ Uses of Sensitivity Analysis

- ◎ There are four major uses of parameter sensitivity analysis.



Validation



Research Design



System Control



Theory



9.2 UNCERTAINTY ANALYSIS

- ◎ Uses of Sensitivity Analysis - Validation
 - ◎ Two different interpretations of sensitivity results pertain to our general judgments of model quality. First, we have an intuitive belief that most real systems will not respond violently to small changes in the values of the operating parameters or variables.
 - ◎ Second, if we are relatively unconfident of the accuracy with which we have estimated a particular parameter and if the model is sensitive to a small change in that parameter, then we should not be confident in the accuracy of the model output.



9.2 UNCERTAINTY ANALYSIS

- ◎ Uses of Sensitivity Analysis – Research Design
 - ◎ Model response will be sensitive to some parameters and not to others.
 - ◎ The sensitive parameters are those to which we should devote the greatest research effort so as to obtain the best estimates, given budget and time constraints.



9.2 UNCERTAINTY ANALYSIS

- ◎ Uses of Sensitivity Analysis – System Control
 - ◎ To control a system means that by altering parameters and variables we can produce desirable output.
 - ◎ If varying a parameter does not alter system output (i.e., the system is insensitive to the parameter), then that parameter is not useful for control.
 - ◎ Therefore, sensitivity analysis can be used to identify which parameters have potential as controllers.



9.2 UNCERTAINTY ANALYSIS

- ◎ Uses of Sensitivity Analysis – Theory
 - ◎ Often the model objective is to investigate a theoretical concept (e.g., conditions for system stability). The response of model output to different parameters may become the central question.
 - ◎ Interesting theoretical questions are to determine which equations can show a particular model behavior and which parameters are responsible for it.



9.2 UNCERTAINTY ANALYSIS

◎ Sensitivity Variables

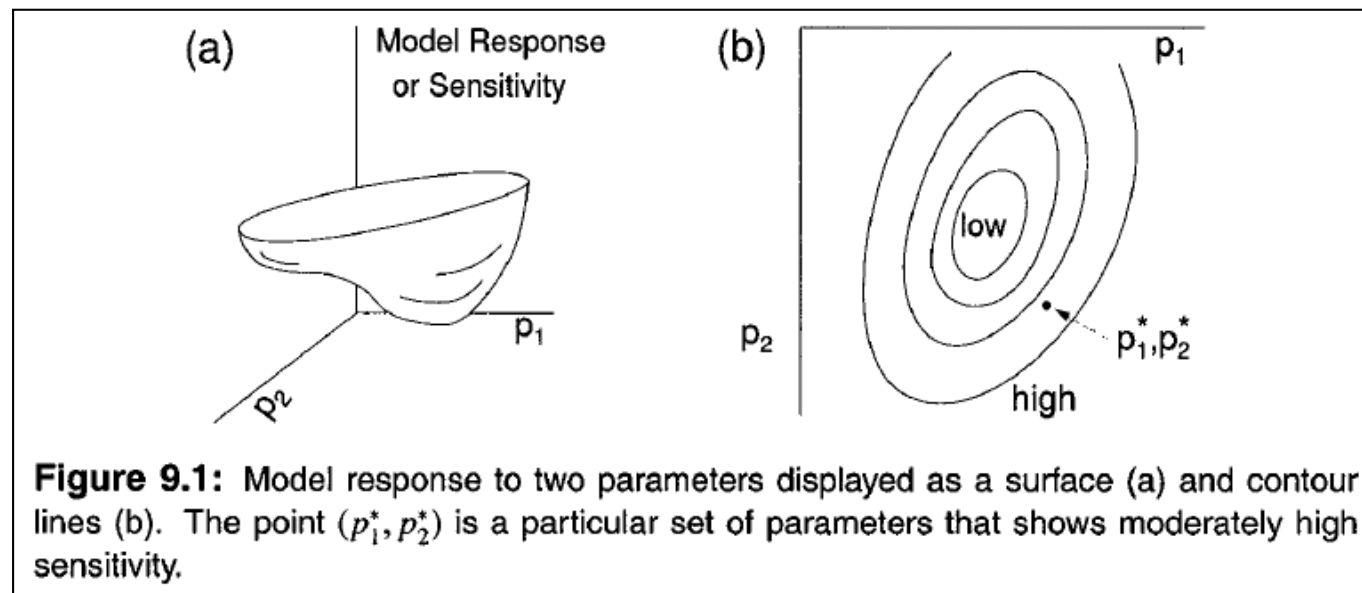
- ◎ The state variables at one or more fixed times.
- ◎ Time averages of state variables.
- ◎ Extreme values (e.g., maximum or minimum) of state variables over a run.
- ◎ Times within a run at which significant events (e.g., extreme values) occur.
- ◎ Simple combinations of state variables are also used, for example, sums, ratios.



9.2 UNCERTAINTY ANALYSIS

◎ Sensitivity Analysis Methods

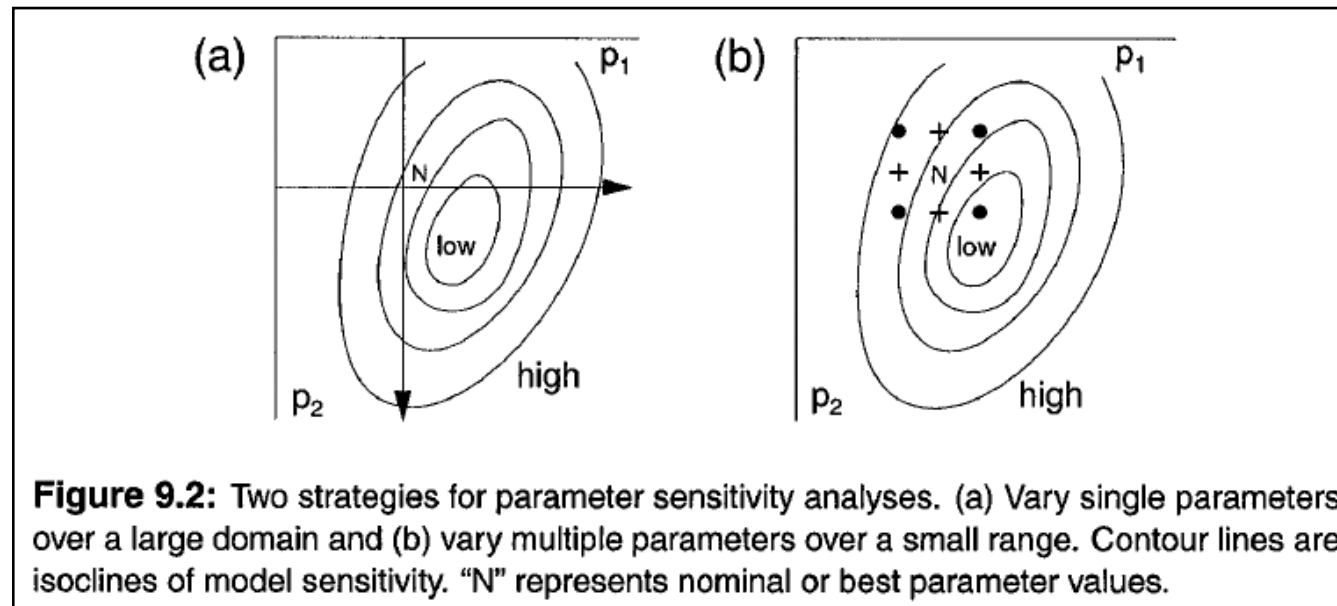
- ◎ When we perform sensitivity analysis we want to answer two questions: (1) How variable is the response? and (2) What are the ranges of model responses to the parameter changes?



9.2 UNCERTAINTY ANALYSIS

◎ Sensitivity Analysis Methods

- ◎ To determine the nature of the surface, sensitivity analysis involves performing numerical experiments in which parameters are systematically changed and resultant model response analyzed.



9.2 UNCERTAINTY ANALYSIS

◎ Single Parameter Sensitivity

- ◎ Sensitivity index S is the ratio of the standardized change in model response (output) to the standardized change in parameter values (input).

$$S = \frac{(R_a - R_n) / R_n}{(P_a - P_n) / P_n} \quad (9.1)$$

- ◎ The question of which parameters and the degree of alteration to study is dependent on the objectives and the purposes of the sensitivity analysis.



9.2 UNCERTAINTY ANALYSIS

◎ Single Parameter Sensitivity

- ◎ There are two strategies for determining the amount by which parameters are altered. In the *uniform approach*, all parameters are altered by the same percentage of the nominal values. The *variable approach* weights the altered interval by the variance of the parameter estimates, if this is known.

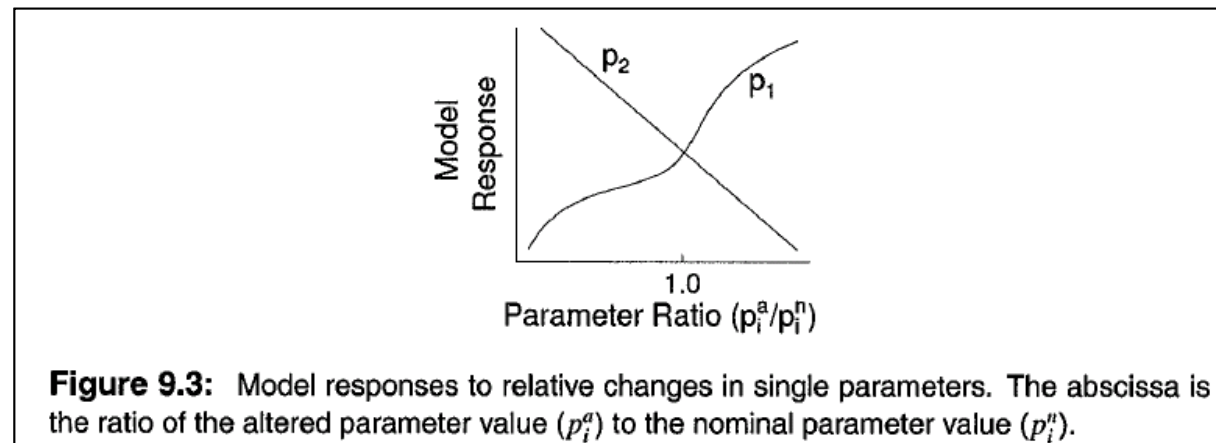
Table 9.1: Sensitivity of density-independent growth to r .

Parm	Nominal input	Nominal output	Altered input	Altered output	S
r^+	0.1	5.4	0.12	6.6	1.15
r^-	0.1	5.4	0.08	4.5	0.88



9.2 UNCERTAINTY ANALYSIS

- ◎ Single Parameter Sensitivity
 - ◎ Normally, we are interested in more than a single parameter, so this table would have additional entries. Also, we are typically interested in more than one response variable.
 - ◎ A graphical presentation showing actual model response (not sensitivity, Fig. 9.3) can be more informative than Table 9.1.



9.2 UNCERTAINTY ANALYSIS

◎ Multiple Parameter Sensitivity

- ◎ For multiple parameter sensitivity, the numerator of Eq. 9.1 does not change, but we must replace the denominator by a distance measure that works in multiple dimensions:

$$d = \sqrt{(p_1 - p'_1)^2 + (p_2 - p'_2)^2}$$

- ◎ An alternative that summarizes the responses of multiple variables over time is:

$$S = \sum_i \sum_j S_{ij}$$

where i indexes time and j indexes the variable.



9.2 UNCERTAINTY ANALYSIS

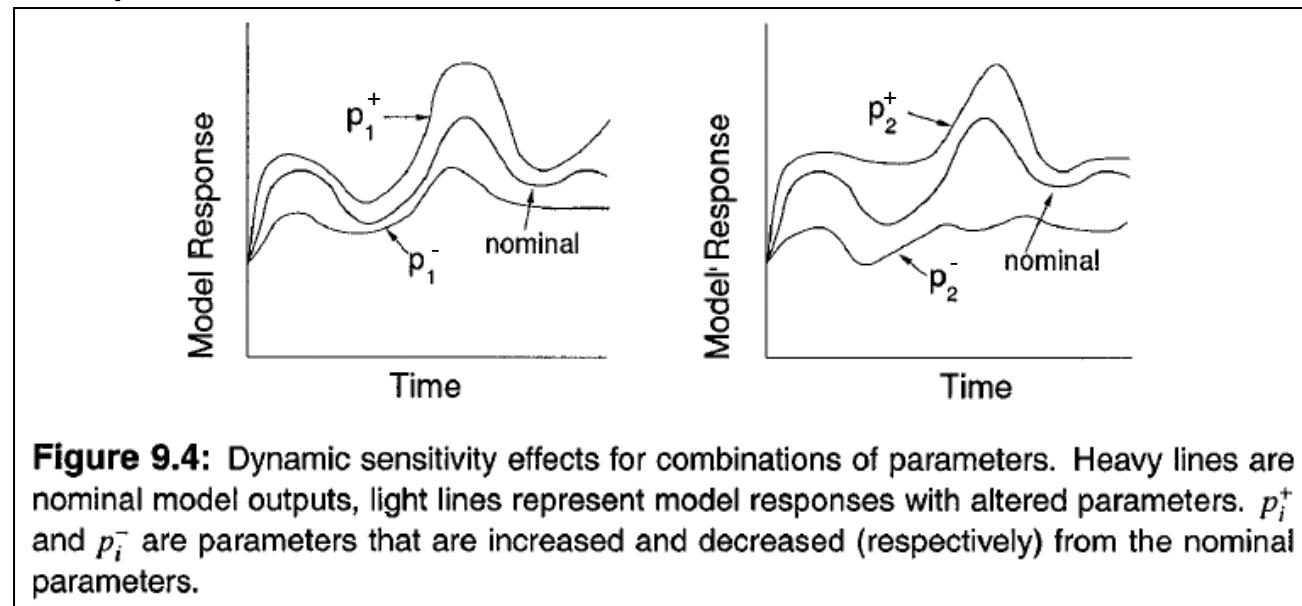
- ◎ Multiple Parameter Sensitivity
 - ◎ An alternative approach is based on a fractional factorial. This approach treats parameter sensitivity analysis as if it were an experimental design for a statistical analysis of empirical data (ANOVA).
 - ◎ The primary sensitivity index is not S , but the F statistic that is computed for analyses of variance. This is used only as a convenient index, and not as a variable for formal hypothesis testing, as it is in true ANOVA.
 - ◎ With this approach, we can gain much information based on relatively few experiments (simulation runs).



9.2 UNCERTAINTY ANALYSIS

◎ Dynamics of Sensitivity

- ◎ It is useful to display the altered model behavior over time. One can also graph the dynamics of the sensitivity index rather than the actual model response.



9.2 UNCERTAINTY ANALYSIS

◎ Error Analysis

- ◎ Whereas sensitivity analysis is concerned with the effects of model response to small changes in the mean parameter values, the author interpret error analysis to be concerned with changes of model response due to the variance of the parameter values.
- ◎ The propagation of error through a function evaluated at a point in the domain depends on the function and the evaluation point. It can be *amplified or compensated*.



9.2 UNCERTAINTY ANALYSIS

◎ Error Analysis

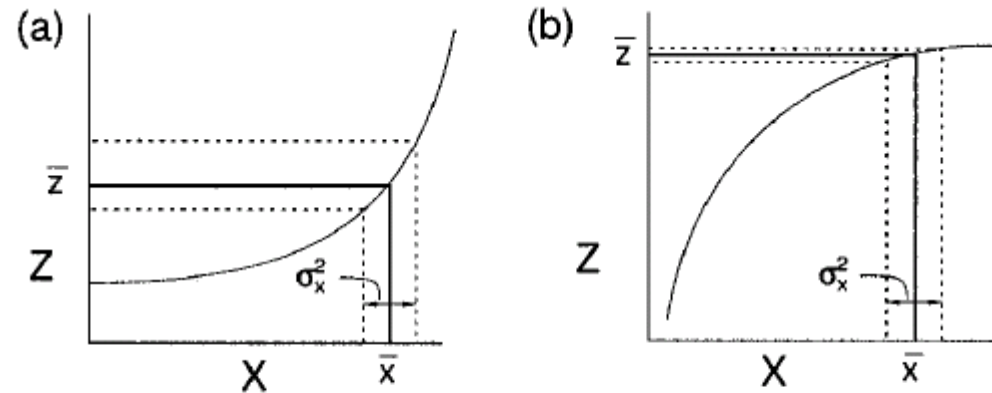


Figure 9.5: Error propagation in simple functions when there is error (σ^2) around the mean of the independent variable ($\bar{X}$). Depending on the function and the mean of the independent variable (X), the error may be *amplified* (a) or *compensated* (b).



9.2 UNCERTAINTY ANALYSIS

◎ Error Analysis

- ◎ The error around functions can be calculated based on the Taylor series expansion of a function in the neighborhood of a point.

$$f(x) = f(a) + \frac{\partial f(x)}{\partial x}(x-a) + \frac{\partial^2 f(x)}{\partial x^2} \frac{(x-a)^2}{2!} + \frac{\partial^3 f(x)}{\partial x^3} \frac{(x-a)^3}{3!} + \dots, \quad (9.2)$$

- ◎ Multivariable form of the Taylor series for functions of more than one independent variable

$$f(x, y, z) \approx f(a, b, c) + \frac{\partial f(x, y, z)}{\partial x}(x-a) + \frac{\partial f(x, y, z)}{\partial y}(y-b) + \frac{\partial f(x, y, z)}{\partial z}(z-c)$$



9.2 UNCERTAINTY ANALYSIS

◎ Error Analysis

- ◎ The first-order Taylor series is the approach we take for developing error propagation equations, the use of which we will call *analytical error analysis*.

$$\langle z \rangle = \bar{z} \approx f(\bar{x}, \bar{y}) + \frac{\partial f}{\partial x}(x - \bar{x}) + \frac{\partial f}{\partial y}(y - \bar{y})$$

The expected value of the function is

$$\bar{z} \approx f(\bar{x}, \bar{y})$$

$$\begin{aligned} \langle (z - \bar{z}) \rangle^2 &= \left(f(\bar{x}, \bar{y}) + \frac{\partial f}{\partial x} \langle (x - \bar{x}) \rangle + \frac{\partial f}{\partial y} \langle (y - \bar{y}) \rangle - f(\bar{x}, \bar{y}) \right)^2 \\ &= \left(\frac{\partial f}{\partial x} \langle (x - \bar{x}) \rangle + \frac{\partial f}{\partial y} \langle (y - \bar{y}) \rangle \right)^2 \\ &= \left(\frac{\partial f}{\partial x} \right)^2 \langle (x - \bar{x}) \rangle^2 + 2 \frac{\partial f}{\partial x} \frac{\partial f}{\partial y} \langle (x - \bar{x})(y - \bar{y}) \rangle + \left(\frac{\partial f}{\partial y} \right)^2 \langle (y - \bar{y}) \rangle^2 \end{aligned}$$



9.2 UNCERTAINTY ANALYSIS

◎ Error Analysis

◎ In general, for n variables

$$\text{var}(z) = \langle (z - \bar{z}) \rangle^2 \approx \sum_{j=1}^n \sum_{i=1}^n \frac{\partial f}{\partial x_i} \frac{\partial f}{\partial x_j} \langle (x_i - \bar{x}_i)(x_j - \bar{x}_j) \rangle \quad (9.3)$$

Table 9.2: Variance formulae for simple functions with correlated and uncorrelated variables.

Function	Uncorrelated	Correlated
$z = x + y$	$\sigma_z^2 = \sigma_x^2 + \sigma_y^2$	$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 + 2\sigma_{xy}$
$z = x - y$	$\sigma_z^2 = \sigma_x^2 + \sigma_y^2$	$\sigma_z^2 = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$
$z = xy$	$\sigma_z^2 = \bar{y}^2 \sigma_x^2 + \bar{x}^2 \sigma_y^2$	$\sigma_z^2 = \bar{y}^2 \sigma_x^2 + \bar{x}^2 \sigma_y^2 + 2\bar{x}\bar{y}\sigma_{xy}$
$z = x/y$	$\sigma_z^2 = \bar{y}^{-2} \sigma_x^2 + \left(\frac{\bar{x}}{\bar{y}^2}\right)^2 \sigma_y^2$	$\sigma_z^2 = \bar{y}^{-2} \sigma_x^2 + \left(\frac{\bar{x}}{\bar{y}^2}\right)^2 \sigma_y^2 - 2\left(\frac{\bar{x}}{\bar{y}^3}\right) \sigma_{xy}$



9.2 UNCERTAINTY ANALYSIS

- ◎ Error Analysis – Analytical Error Analysis
 - ◎ Analytical error analysis uses the error propagation of functions, given that the model can be reduced to a single equation that predicts some quantity of interest.
 - ◎ Uncertainties of the true values of the parameters will propagate to create uncertainties of the predicted probability.



9.2 UNCERTAINTY ANALYSIS

- ◎ Error Analysis – Analytical Error Analysis
 - ◎ A simple example model of the probability of extinction due to demographic stochasticity (Pielou, 1977):

$$P = \left(\frac{d}{b} \right)^n \quad (9.4)$$

Applying Eq. 9.3 to Eq. 9.4 to estimate the prediction uncertainty

$$\text{var}(P) = \left(\frac{\bar{n} \bar{d}^{\bar{n}-1}}{\bar{b}^{\bar{n}}} \right)^2 \sigma_d^2 + \left(\frac{\bar{n} \bar{d}^{\bar{n}} \bar{b}^{\bar{n}-1}}{\bar{b}^{2\bar{n}}} \right)^2 \sigma_b^2 + \left(\ln \left(\frac{\bar{d}}{\bar{b}} \right) \left(\frac{\bar{d}}{\bar{b}} \right)^{\bar{n}} \right)^2 \sigma_n^2 \quad (9.5)$$



9.2 UNCERTAINTY ANALYSIS

- ◎ Error Analysis – Analytical Error Analysis
 - ◎ Suppose we have these values for means and standard deviations:

	d	b	n
mean	0.8	0.9	10
std. dev.	0.157	0.174	0.69

Then, the expected $P = 0.308$; the variance is $\text{var}(P) = 0.72$; and the standard deviation is 0.849. The 95% confidence intervals around the mean is

$$CI_{lower} = 0.308 - (1.96)(0.849) = -1.356$$

$$CI_{upper} = 0.308 + (1.96)(0.849) = 1.972$$



9.2 UNCERTAINTY ANALYSIS

- ◎ Error Analysis – Monte Carlo Error Analysis
 - ◎ Error analysis using Monte Carlo techniques (Chapter 10) can be applied to complex dynamic models and do not require extensive mathematical analysis.
 - ◎ The method is to simulate repeatedly a system of equations using randomly selected parameter values. The output of each run is collected and statistically analyzed after all runs have been performed.
 - ◎ Individual parameter values are selected from frequency distributions appropriate for each parameter.



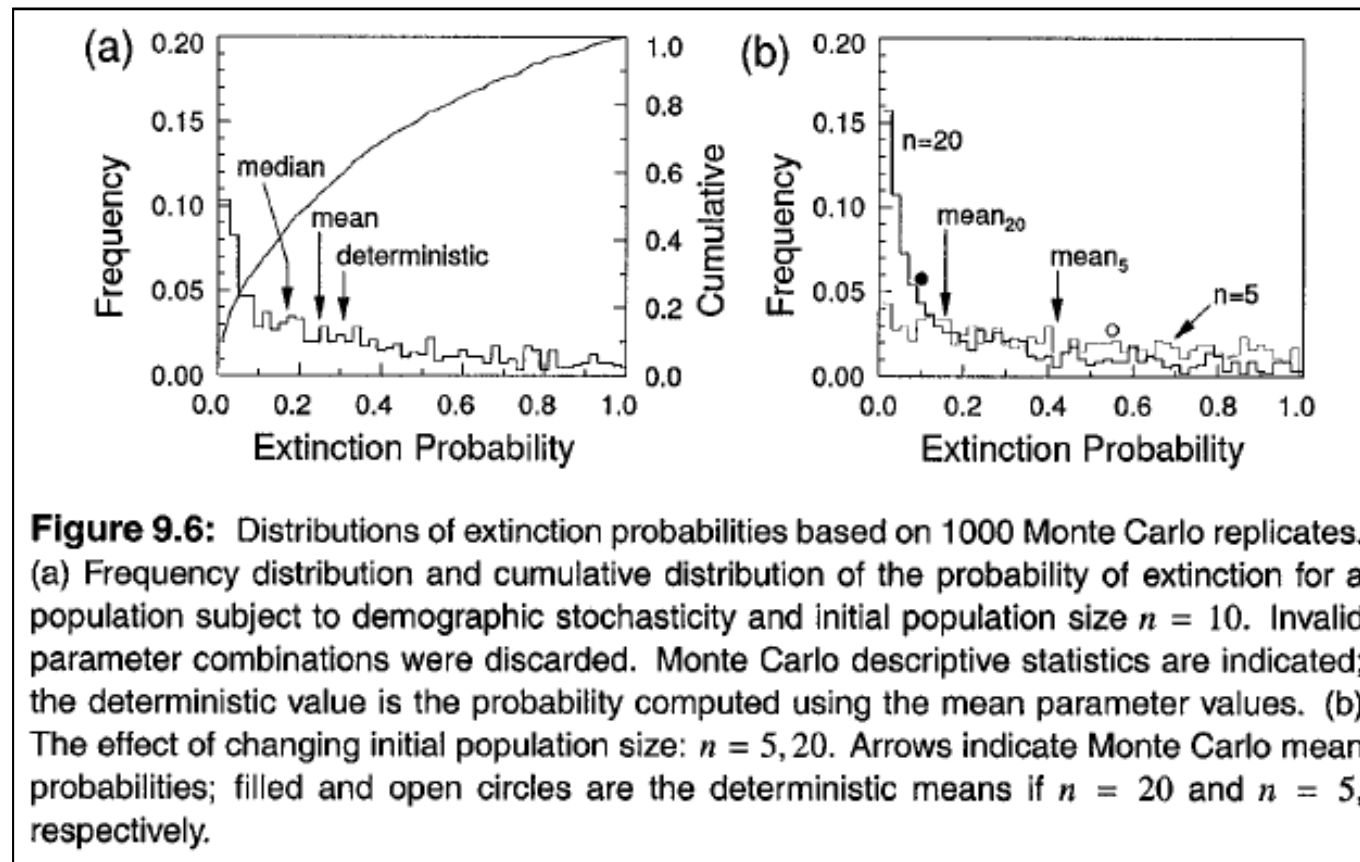
9.2 UNCERTAINTY ANALYSIS

- ◎ Error Analysis – Monte Carlo Error Analysis
 - ◎ Two important practical problems arise when implementing a Monte Carlo analysis of error. First, we must decide what probability distribution from which to choose the parameters.
 - ◎ If the parameter distribution is unknown, another problem is to choose a distribution that is consistent with basic biological knowledge.
 - ◎ The second practical concern is to ensure that our scheme for sampling from the probability distribution(s) adequately represents the tails of the distributions.



9.2 UNCERTAINTY ANALYSIS

Error Analysis – Monte Carlo Error Analysis



9.2 UNCERTAINTY ANALYSIS

◎ Aggregation Analysis

- ◎ We wish our models to be as simple as possible. One approach is to maintain a high degree of biological detail but curtail the extent of the system. Another approach to simplification is to lump system variables together.
- ◎ This strategy of lumping variables is known as "aggregation," and we want to estimate the errors that aggregation introduces into model output.
- ◎ In practice, all models are aggregated at some level of biological organization.
- ◎ An aggregated model is one in which the state variables of a more detailed model are lumped to form a subset.



9.2 UNCERTAINTY ANALYSIS

◎ Aggregation Analysis

- ◎ Iwasa et al. (1987) developed an aggregation theory based on a restrictive definition. *Perfect aggregation*, by their definition, is an aggregation that produces identical dynamics at each point of time considered.
- ◎ Additional progress has been made with a more relaxed attitude toward dynamic similarity. One such relaxation is that the equilibrium of the sum of the state variables of the detailed model should equal the equilibrium of the aggregated model.
- ◎ There are few analytical tools to assess the amount of error that is made by our choice of state variables. This leaves us with Monte Carlo simulation of particular cases as the main tool to unravel errors that arise from lumping state variables.



9.2 UNCERTAINTY ANALYSIS

- ◎ Uncertainty Analysis and Validation
 - ◎ Sensitivity analysis, error analysis, and aggregation analysis, are all methods that allow us to explore model behavior in different regions of “prediction space”.
 - ◎ Combining these comparisons with empirical studies of a wide range of real systems, so that we can assess both model reliability and model adequacy, will go a long way towards increasing our confidence (or lack of confidence) in the myriad models now being produced of extremely complex systems.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Equilibria

- ◎ A system of differential or difference equations is in equilibrium if the values of the state variables are not changing in time. Equilibrium analysis seeks to identify the values of all the equilibria.
- ◎ Knowing the equilibria of a model is useful for several reasons:
 - First, it characterizes the long-term behavior of the model by providing a set of algebraic equations that depend on the parameters and state variables.
 - Second, knowing the location and number of equilibria for a model can help us interpret the transient dynamics that we observe from simulation.
 - Third, the equilibria are the points at which we discuss the stability properties of the model.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Equilibria – Example of Lotka-Volterra Equations

$$\frac{dV}{dt} = aV - bVP$$

$$\frac{dP}{dt} = cbVP - dP$$

Solving for the equilibria gives

$$0 = aV^* - bV^*P^* \quad (9.7)$$

$$0 = cbV^*P^* - dP^* \quad (9.8)$$

Solving for V^* and P^*

$$P^* = a / b \quad V^* = d / cb \quad (9.9)$$

$$V^* = 0 \quad P^* = 0 \quad (9.10)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

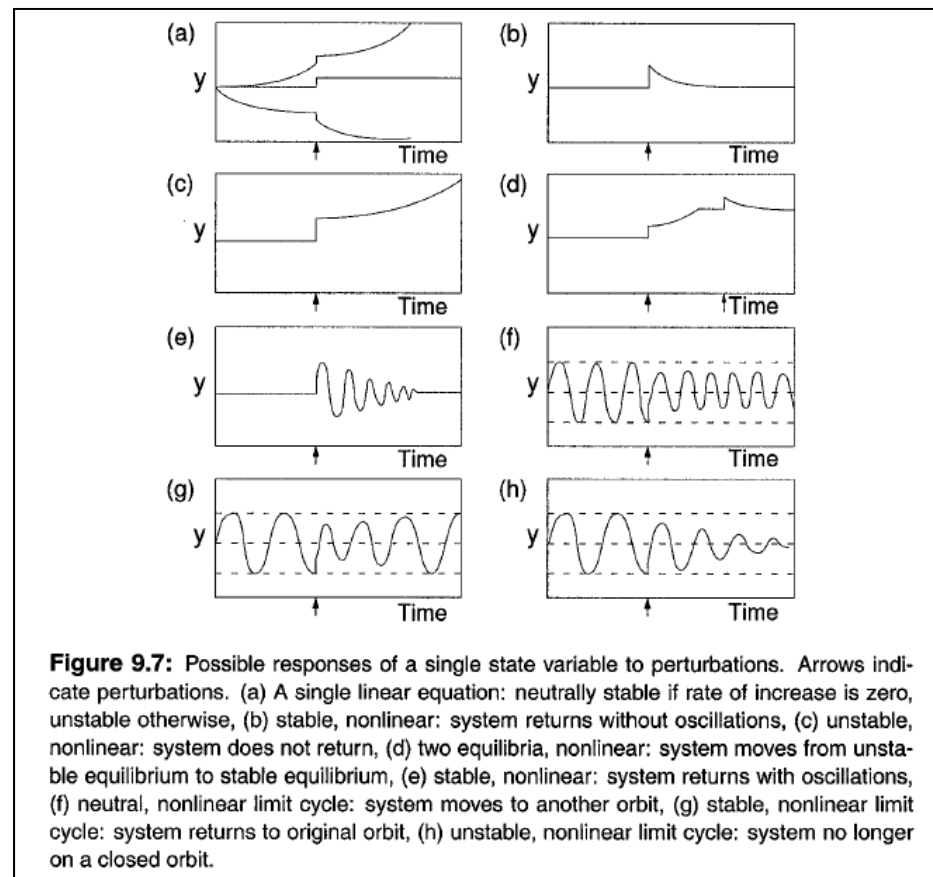
◎ Stability: The Concept

- ◎ Stability analysis is the analysis of a system of differential equations to determine the dynamics over short times of the system in response to small perturbations.
- ◎ Intuitively, a system is stable following a perturbation of one or more of the state variables if the system returns to the specific point in state space or to a specific orbit (trajectory) in state space.
- ◎ In general, we are interested in the global response of the system to perturbations. This is difficult for many nonlinear systems, and we usually are able to complete only a local (or neighborhood) analysis.



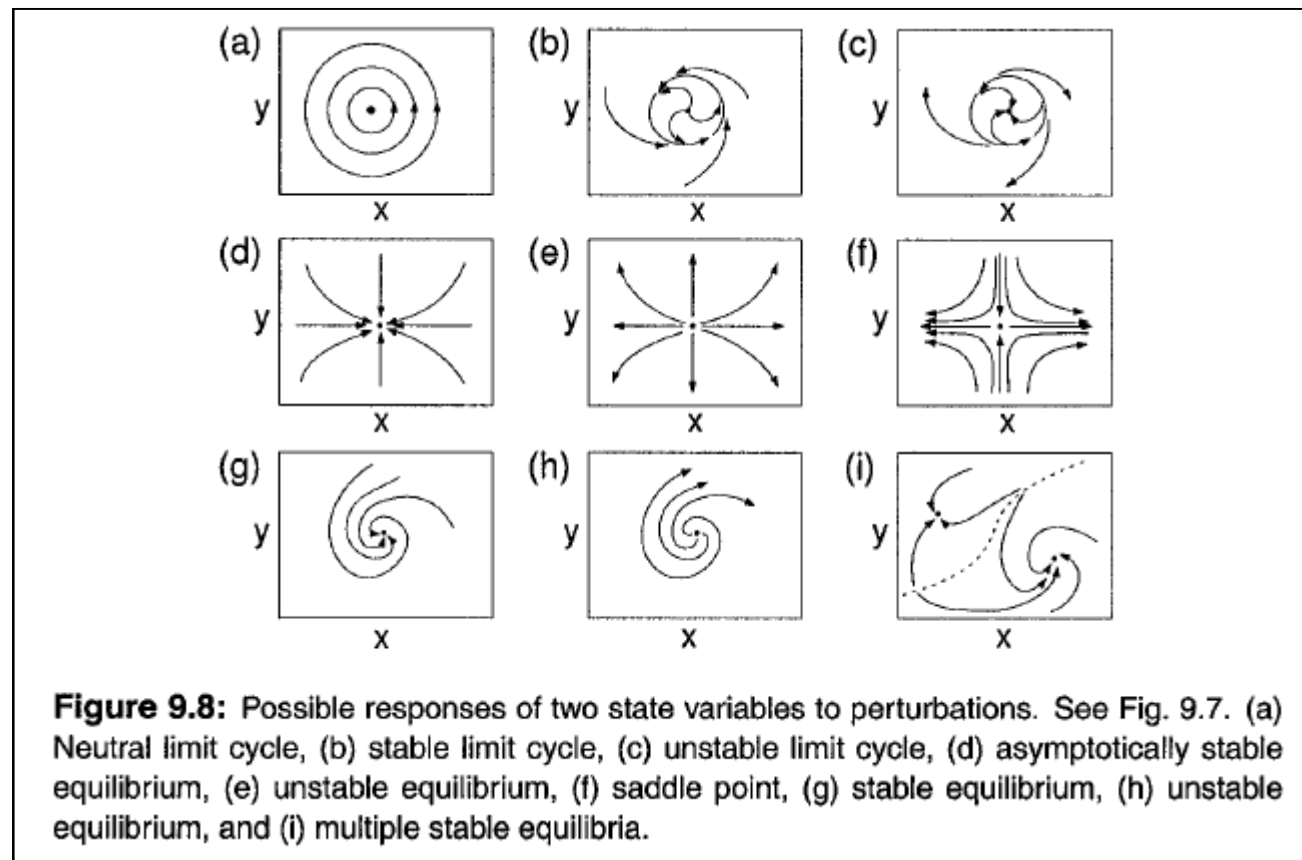
9.3 ANALYSIS OF MODEL BEHAVIOR

◎ A Menagerie of System Responses



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ A Menagerie of System Responses



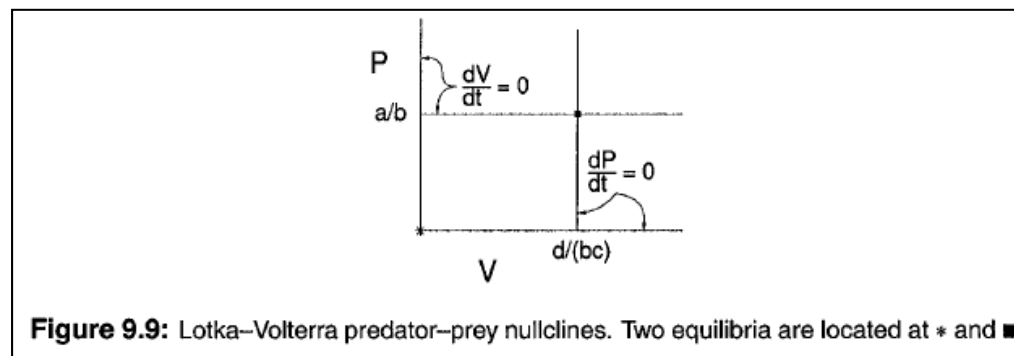
9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nullclines and Graphical Stability

- ◎ The *nullclines* of a system of differential equations are the set of points in state space that satisfy the equilibria equations for each of the state variables.
- ◎ For Lotka-Volterra model (Eqs. 9.7 and 9.8), there are two nullclines for each state variable.

$$0 = aV - bVP \quad \leftarrow \text{Victim equilibria} \quad (9.11)$$

$$0 = cbVP - dP \quad \leftarrow \text{Predator equilibria} \quad (9.12)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nullclines and Graphical Stability

◎ The two-species Gause Competition equations

$$\frac{dn_1}{dt} = r_1 n_1 \left(1.0 - \left(\frac{n_1 + \alpha n_2}{K_1} \right) \right) = \underbrace{r_1 n_1}_{\text{unrestricted growth}} - \underbrace{\frac{r_1 n_1 n_1}{K_1}}_{\text{intra-specific competition}} - \underbrace{\frac{r_1 n_1 (\alpha n_2)}{K_1}}_{\text{inter-specific competition}} \quad (9.13)$$

$$\frac{dn_2}{dt} = r_2 n_2 \left(1.0 - \left(\frac{n_2 + \beta n_1}{K_2} \right) \right) = r_2 n_2 - \frac{r_2 n_2 n_2}{K_2} - \frac{r_2 n_2 (\beta n_1)}{K_2} \quad (9.14)$$

Four nullclines can be obtained:

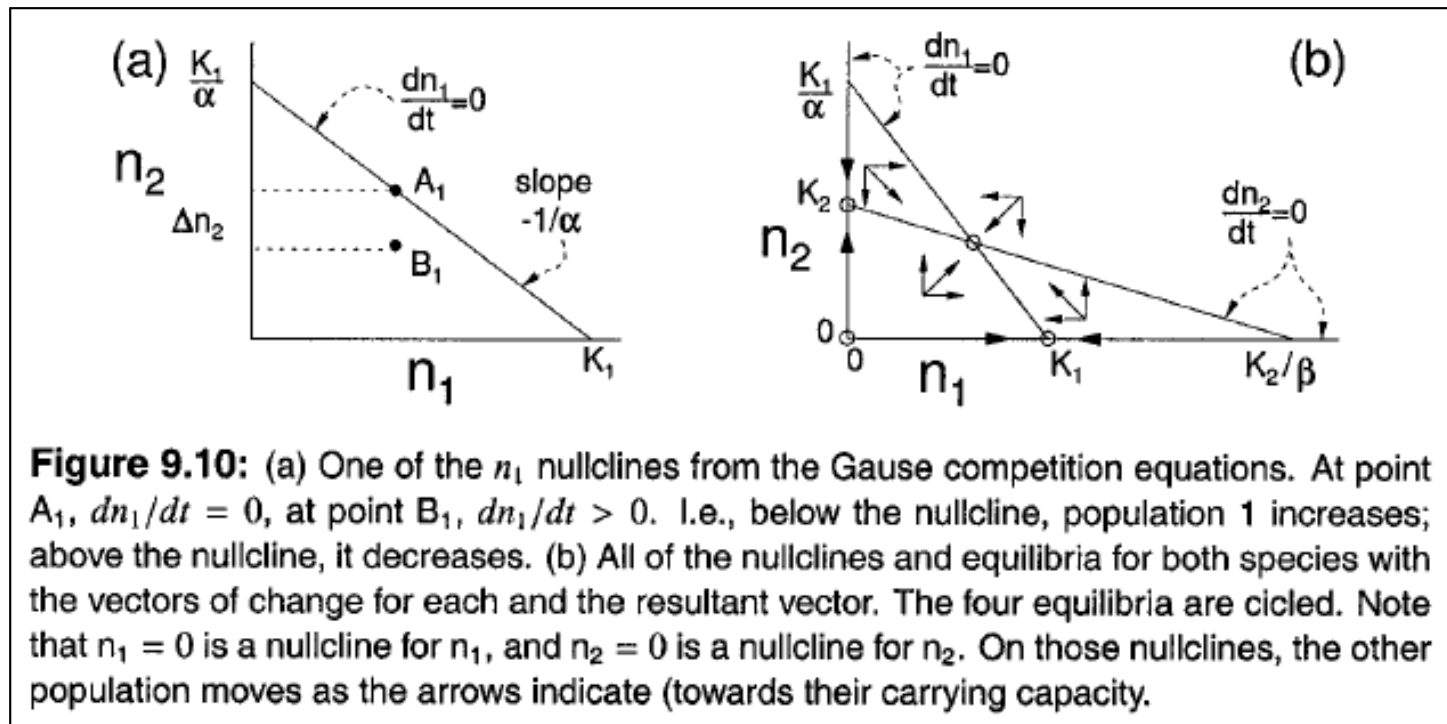
$$n_1 = K_1 - \alpha n_2 \quad \text{and} \quad n_1 = 0 \quad \leftarrow n_1 \text{ nullclines} \quad (9.15)$$

$$n_2 = K_2 - \beta n_1 \quad \text{and} \quad n_2 = 0 \quad \leftarrow n_2 \text{ nullclines} \quad (9.16)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

- ◎ Nullclines and Graphical Stability
 - ◎ The two-species Gause Competition equations



9.3 ANALYSIS OF MODEL BEHAVIOR

- ◎ Nullclines and Graphical Stability
 - ◎ The two-species Gause Competition equations

Determination of equilibria :

$$(0,0), (K_1,0),(0,K_2),(n_1^*,n_2^*)$$

$$\begin{aligned}n_2^* &= K_2 - \beta(K_1 - \alpha n_2^*) && \leftarrow \text{Substitute Eq. 9.15 into Eq. 9.16} \\&= K_2 - \beta K_1 + \alpha \beta n_2^* \\&= \frac{K_2 - \beta K_1}{1 - \alpha \beta} && \leftarrow \text{Factor} \quad (9.17)\end{aligned}$$

$$n_1^* = \frac{K_1 - \alpha K_2}{1 - \alpha \beta} \quad \leftarrow \text{Substitute } n_2^* \text{ into Eq. 9.15} \quad (9.18)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nullclines and Graphical Stability

◎ The two-species Gause Competition equations

The rates of change of the population at A_1 and at B_1 are:

$$\begin{aligned}\left(\frac{dn_1}{dt}\right)_{A_1} &= r_1 n_{1A} \left(1.0 - \left(\frac{n_{1A} + \alpha n_{2A}}{K_1}\right)\right) = 0 \\ \left(\frac{dn_1}{dt}\right)_{B_1} &= r_1 n_{1A} \left(1.0 - \left(\frac{n_{1A} + \alpha(n_{2A} - \Delta n_2)}{K_1}\right)\right)\end{aligned}\quad (9.19)$$

Rearranging Eq. 9.19 as

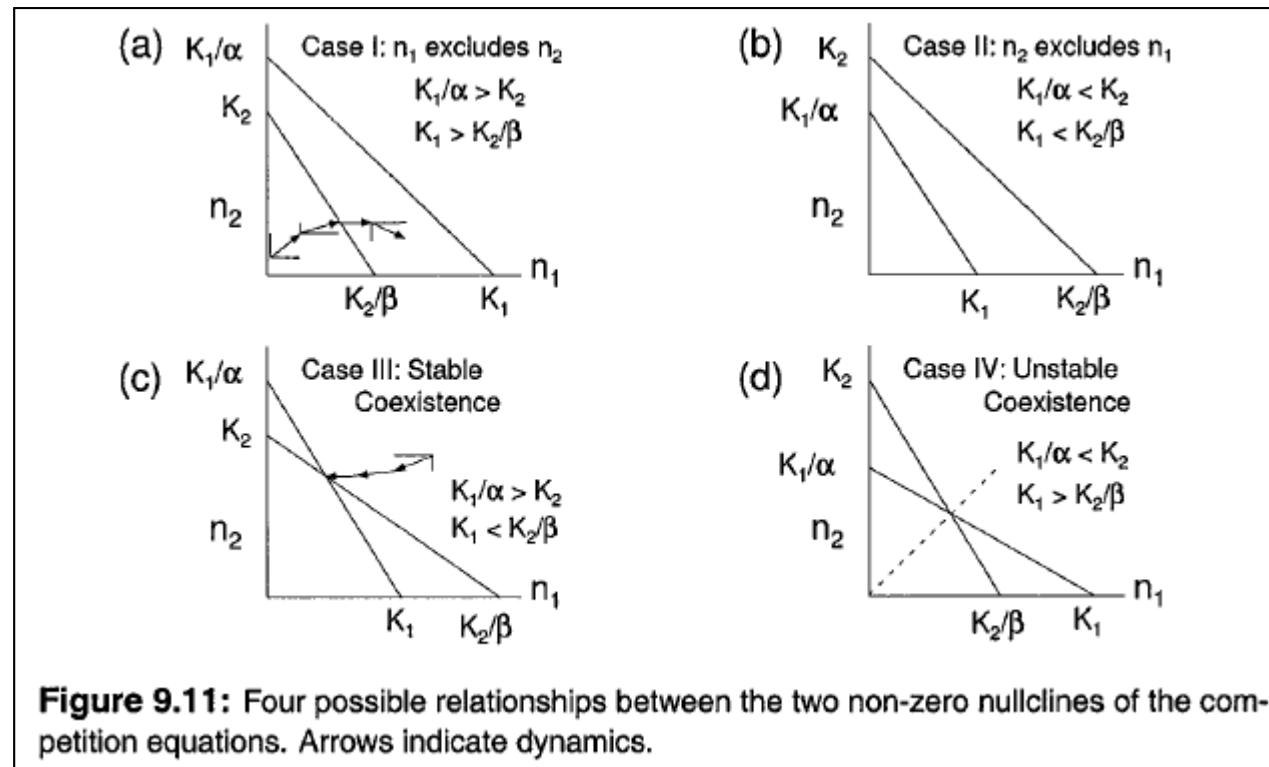
$$\begin{aligned}\left(\frac{dn_1}{dt}\right)_{B_1} &= r_1 n_{1A} \left(1.0 - \left(\frac{n_{1A} + \alpha n_{2A}}{K_1}\right) + \frac{\alpha \Delta n_2}{K_1}\right) \\ &= \left(\frac{dn_1}{dt}\right)_{A_1} + r_1 n_{1A} \frac{\alpha \Delta n_2}{K_1} = r_1 n_{1A} \frac{\alpha \Delta n_2}{K_1}\end{aligned}\quad (9.20)$$

$$\text{Thus } \left(\frac{dn_1}{dt}\right)_{B_1} > \left[\left(\frac{dn_1}{dt}\right)_{A_1} = 0\right]$$



9.3 ANALYSIS OF MODEL BEHAVIOR

- ◎ Nullclines and Graphical Stability
- ◎ The two-species Gause Competition equations



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

- ◎ To address stability in nonlinear models, we must linearize the equations at the equilibria, then use the techniques that apply for linear equations.
- ◎ The linear approximation is valid only for small regions around the equilibrium.
- ◎ Consider the standard linear model in population ecology, the density-independent growth equation:

$$\frac{dN}{dt} = rN \quad (9.21)$$

$$N_t = N_0 e^{rt} \quad (9.22)$$

If $r > 0$, the system will move away from the equilibrium and will be *unstable*. Otherwise, the system will be *stable*.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

- ◎ An example of a linear, two-state variable model:

$$\begin{pmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix} \quad (9.23)$$

$$x_1 = c_1 e^{\lambda t} \quad x_2 = c_2 e^{\lambda t} \quad (9.24)$$

- ◎ Track 1: Derivative of the Proposed Solution

$$\frac{dx_1}{dt} = \lambda c_1 e^{\lambda t} \quad (9.25)$$

$$\frac{dx_2}{dt} = \lambda c_2 e^{\lambda t} \quad (9.26)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

◎ Track 2: Insert Proposed Solution into Differential Equations

$$\begin{aligned}\frac{dx_1}{dt} &= a_{11}x_1 + a_{12}x_2 = a_{11}(c_1e^{\lambda t}) + a_{12}(c_2e^{\lambda t}) \\ &= e^{\lambda t}(a_{11}c_1 + a_{12}c_2)\end{aligned}\tag{9.27}$$

$$\frac{dx_2}{dt} = e^{\lambda t}(a_{21}c_1 + a_{22}c_2)\tag{9.28}$$

◎ Combining Tracks 1 and 2

$$\lambda c_1 = a_{11}c_1 + a_{12}c_2 \quad \text{(from Eqs. 9.25 and 9.27)}$$

$$\lambda c_2 = a_{21}c_1 + a_{22}c_2 \quad \text{(from Eqs. 9.26 and 9.28)}$$



9.3 ANALYSIS OF MODEL BEHAVIOR

- ◎ Linear System Stability Analysis
 - ◎ Combining Tracks 1 and 2

$$\lambda \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$\lambda \mathbf{c} = \mathbf{A} \mathbf{c} \quad (9.29)$$

$$\lambda \mathbf{c} - \mathbf{A} \mathbf{c} = 0 \quad (9.30)$$

$$(\mathbf{A} - \lambda \mathbf{I}) \mathbf{c} = 0 \quad (9.31)$$

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0 \quad (9.32)$$

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0$$

$$\lambda^2 - (a_{11} + a_{22})\lambda + a_{11}a_{22} - a_{12}a_{21} = 0$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

◎ Interpreting λ

$$\begin{aligned} x_1 &= c_1 e^{\lambda t} \\ x_2 &= c_2 e^{\lambda t} \end{aligned} \tag{9.33}$$

$$\begin{aligned} x_1 &= \alpha_1 c_{11} e^{\lambda_1 t} + \alpha_2 c_{12} e^{\lambda_2 t} \\ x_2 &= \alpha_1 c_{21} e^{\lambda_1 t} + \alpha_2 c_{22} e^{\lambda_2 t} \end{aligned} \tag{9.34}$$

The vector $\mathbf{c}_1 = [c_{11}, c_{21}]$ and $\mathbf{c}_2 = [c_{12}, c_{22}]$ are the eigenvectors associated with λ_1 and λ_2 , respectively.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

For real λ

- ◎ If the largest λ_i is positive, the time solutions increase; that is, the system dynamics take the perturbation further from the equilibrium point and the system is unstable.
- ◎ If the largest λ_i is negative, the perturbation decreases with time and the system is stable.
- ◎ If the largest λ_i is positive and at least one is negative, we have a saddle (or mountain pass): the system is stable along a finite number of paths (the mountain ridge), but unstable for all other perturbations.
- ◎ If $\lambda = 0$, the perturbation neither increases or decreases, and the system has neutral stability.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

For imaginary λ

$$\begin{aligned} x(t) &= \alpha_1 c_{11} e^{\rho t} e^{i\kappa t} + \alpha_2 c_{12} e^{\rho t} e^{-i\kappa t} \\ y(t) &= \alpha_1 c_{21} e^{\rho t} e^{i\kappa t} + \alpha_2 c_{22} e^{\rho t} e^{-i\kappa t} \end{aligned} \quad (9.35)$$

$$e^{i\kappa t} = \cos(\kappa t) + i \sin(\kappa t)$$

$$\begin{aligned} x(t) &= \alpha_1 e^{\rho t} [c_{11} \cos(\kappa t) - c_{12} \sin(\kappa t)] + \alpha_2 e^{\rho t} [c_{11} \sin(\kappa t) - c_{12} \cos(\kappa t)] \\ y(t) &= \alpha_1 e^{\rho t} [c_{21} \cos(\kappa t) - c_{22} \sin(\kappa t)] + \alpha_2 e^{\rho t} [c_{21} \sin(\kappa t) - c_{22} \cos(\kappa t)] \end{aligned} \quad (9.36)$$

This tells us that the long-term dynamics of a system of linear differential equations without a forcing function will be a sum of sines and cosines. Cycles can be produced by these simple models in two or more dimensions, whereas they could not be produced in systems with a single state variable.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Linear System Stability Analysis

For imaginary λ

- ◎ If $\rho = 0$ and $\kappa \neq 0$, the solution is a sum of a cosine and sine function with constant amplitude. Therefore, a perturbation of the equilibrium will cause undamped oscillations (neutral stability, Fig. 9.8a).
- ◎ If $\rho > 0$, the amplitudes of oscillations grow exponentially and the solution is unstable (Fig. 9.8h).
- ◎ If $\rho < 0$, the oscillations are damped and the solution is stable (Fig. 9.8g).
- ◎ The frequency of the oscillations are determined by the κ_j .



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations

- ◎ We wish to convert a nonlinear equation to a linear equation so that the above neighborhood stability analysis can be performed. Basically, we wish to define a new function of the deviations of the system following perturbation from the equilibrium.
- ◎ To show the linearization method for X_1 , we begin by perturbing the equilibrium point by an amount x_1

$$\frac{d(X_1^* + x_1)}{dt} = f(X_1^* + x_1)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations

$$\frac{d(X^* + x)}{dt} = \underbrace{f(X^*)}_{\text{zero-order}} + x \underbrace{\frac{\partial f}{\partial x}}_{\text{first-order}}$$

It is also true that

$$\frac{d(X^* + x)}{dt} = \frac{dX^*}{dt} + \frac{dx}{dt}$$

So we have the first - order approximation

$$\frac{dx}{dt} = x \frac{\partial f}{\partial x}$$

Therefore, for a system of two ODEs

$$\begin{aligned} \frac{dx}{dt} &= x \left. \frac{\partial f}{\partial x} \right|_{X^*} + y \left. \frac{\partial f}{\partial y} \right|_{X^*} \\ \frac{dy}{dt} &= x \left. \frac{\partial g}{\partial x} \right|_{X^*} + y \left. \frac{\partial g}{\partial y} \right|_{X^*} \end{aligned} \quad (9.37)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations - Example

$$\frac{dx_1}{dt} = f_1(x_1, x_2, x_3) \approx x_1 \frac{\partial f_1}{\partial x_1} + x_2 \frac{\partial f_1}{\partial x_2} + x_3 \frac{\partial f_1}{\partial x_3}$$

$$\frac{dx_2}{dt} = f_2(x_1, x_2, x_3) \approx x_1 \frac{\partial f_2}{\partial x_1} + x_2 \frac{\partial f_2}{\partial x_2} + x_3 \frac{\partial f_2}{\partial x_3}$$

$$\frac{dx_3}{dt} = f_3(x_1, x_2, x_3) \approx x_1 \frac{\partial f_3}{\partial x_1} + x_2 \frac{\partial f_3}{\partial x_2} + x_3 \frac{\partial f_3}{\partial x_3}$$

We can write this set of equations as a matrix

$$\dot{\mathbf{x}} = \mathbf{J}\mathbf{x} \quad (9.38)$$

$$= \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad (9.39)$$



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations - Example

The Gause Eqs. 9.13 and 9.14 have the following Jacobian for deviations from the equilibrium defined by n_1^* and n_2^* :

$$\frac{\partial f_1}{\partial n_1} = r_1 - \frac{2r_1 n_1^*}{K_1} - \frac{r_1 n_2^* \alpha}{K_1} \quad \frac{\partial f_1}{\partial n_2} = -\frac{r_1 n_1^* \alpha}{K_1} \quad (9.40a)$$

$$\frac{\partial f_2}{\partial n_1} = -\frac{r_2 n_2^* \beta}{K_2} \quad \frac{\partial f_2}{\partial n_2} = r_2 - \frac{2r_2 n_2^*}{K_2} - \frac{r_2 n_1^* \beta}{K_2} \quad (9.40b)$$

r_1	α	K_1	r_2	β	K_2
0.05	0.2	200	0.05	6.0	1100



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations – Example

$$\mathbf{J} = \begin{pmatrix} -0.025 & -0.005 \\ -0.1363636 & -0.022772727 \end{pmatrix}$$

$$\begin{vmatrix} (-0.025 - \lambda) & -0.005 \\ -0.1363636 & (-0.022772727 - \lambda) \end{vmatrix} = 0$$

$$\lambda^2 + 0.04772727\lambda - 0.000113636 = 0$$

The roots of the polynomial are

$$\lambda_1 = 0.0460379 \quad \lambda_2 = -0.093765171$$

Since the largest eigenvalue is positive, we conclude the system is not stable. Since λ_2 is negative, we have a saddle point (Fig. 9.8f), that is, a ridge along which the system will converge to the equilibrium. This ridge is sometimes called a *separatrix*.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations

- ◎ Since the roots are determined completely from the coefficients of the polynomial contained in the elements of the Jacobian matrix. It is possible, therefore, to ascertain the sign of the eigenvalue simply by inspecting the constants of the matrix.
- ◎ These relationships have been codified in several stability criteria. Two of the more important of these are the criteria of Routh and Hurwitz.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Nonlinear Equations

- ◎ As an example of the method, consider a general characteristic equation for a system of m state variables:

$$\lambda^m + a_1 \lambda^{m-1} + a_2 \lambda^{m-2} + \cdots + a_m = 0 \quad (9.41)$$

- ◎ A system of two state variables ($m = 2$) will be stable if and only if $a_1 > 0$ and $a_2 > 0$. If $m = 3$, then the system will be stable if and only if $a_1 > 0$, $a_3 > 0$, and $a_1 a_2 > a_3$. With these rules, stability can be determined without actually having to find the roots of a polynomial. May (1973) lists the rules for $m = 1, \dots, 5$.



9.3 ANALYSIS OF MODEL BEHAVIOR

- ◎ Displaying Stability Analyses
 - ◎ Stability diagrams are (usually) two dimensional graphs each axis of which is a parameter (or combination of parameters) that are chosen because they are important in controlling the system's stability properties. In the graph, lines are drawn demarcating regions of this parameter space that have different stability properties (e.g., "stable-equilibrium," "limit-cycle", etc). Examples of these can be seen in Figs. 14.7, 18.5, 18.20.
 - ◎ The second method of displaying system stability behavior is by graphing multiple nullclines on a single plot. Examples of this technique can be found in Figs. 11.5, 11.6, and 13.10.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Précis on Stability Analysis

Steps in Doing a Neighborhood Stability Analysis

1. Determine equilibria for particular parameters.
2. If nonlinear, compute the Jacobian matrix.
3. Create the characteristic equation and compute eigenvalues.
4. Inspect the real part of λ_i : $\max(\text{Re } \lambda_i) < 0$ implies stability.
5. Or, use the Routh-Hurwitz criteria.



9.3 ANALYSIS OF MODEL BEHAVIOR

◎ Précis on Stability Analysis

- ◎ Stability analysis is an elegant, but limited, tool.
- ◎ It is not always possible to find a closed form solution for the equilibria (even if it does exist in the model). Nullcline analysis addresses the same questions, but graphically. It has great heuristic power, but is difficult to perform for more than three state variables.
- ◎ Overall, stability analysis is one of several tools available for understanding model behavior to be used where appropriate.



9.4 MATHEMATICAL DETAILS

◎ Why Equation 9.32 Must Be True

$$d_{11}x_1 + d_{12}x_2 = b_1 \quad (9.42a)$$

$$d_{21}x_1 + d_{22}x_2 = b_2 \quad (9.42b)$$

$$\begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$-\frac{d_{21}}{d_{11}}(d_{11}x_1 + d_{12}x_2) = -\frac{d_{21}}{d_{11}}b_1$$

$$-d_{21}x_1 - \frac{d_{21}d_{12}}{d_{11}}x_2 = -\frac{d_{21}}{d_{11}}b_1$$

$$\text{add to Eq. 9.42b to get: } 0 + d_{22}x_2 - \frac{d_{21}d_{12}}{d_{11}}x_2 = b_2 - \frac{d_{21}}{d_{11}}b_1$$



9.4 MATHEMATICAL DETAILS

Why Equation 9.32 Must Be True

$$x_2 \left(\frac{d_{22}d_{11} - d_{21}d_{12}}{d_{11}} \right) = \frac{d_{11}b_2 - d_{21}b_1}{d_{11}}$$
$$x_2 = \frac{d_{11}b_2 - d_{21}b_1}{d_{22}d_{11} - d_{21}d_{12}} \quad (9.43)$$

Substitute Eq. 9.43 into Eq. 9.42b and solve for x_1 :

$$x_1 = \frac{b_1d_{22} - b_2d_{12}}{d_{22}d_{11} - d_{21}d_{12}} \quad (9.44)$$

The solution of n equations and n unknowns is the ratio of determinants, the denominator of which is the determinant of the original matrix of coefficients. This is a very deep result; it is known as Cramer's Rule.



9.4 MATHEMATICAL DETAILS

☉ Why Equation 9.32 Must Be True

$$(\mathbf{A} - \lambda \mathbf{I})\mathbf{c} = \mathbf{0} \quad (9.45)$$

$$\mathbf{x} = \frac{\mathbf{0}}{\det(\mathbf{d})}$$

Defining the matrix $\mathbf{d}$ for the stability problem and re - arranging:

$$\det(\mathbf{d})\mathbf{x} = \mathbf{0}$$

$$\det(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0} \quad (9.46)$$

The problem was to find the solution to a system of n equations with n unknowns, which yields Eq. 9.46. Either all the x_i in that equation are 0 (trivial solution), or, in the non - trivial case of interest, $\det(\mathbf{A} - \lambda \mathbf{I})\mathbf{x} = \mathbf{0}$.



9.4 MATHEMATICAL DETAILS

⊙ Eigenvectors

- ⊙ An eigenvector is an n -valued vector, where n is the dimension, or number of state variables. There is one eigenvector associated with each eigenvalue. The elements of the eigenvector $\mathbf{c}$ satisfy

$$\lambda \mathbf{c} = \mathbf{A} \mathbf{c}$$

or

$$(\mathbf{A} - \lambda_i \mathbf{I}) \mathbf{c}_i = 0 \quad (9.47)$$

- ⊙ The eigenvalue determines how steeply its exponential increases or decreases in t timesteps. The elements of the eigenvector determine the relative distribution of that change over the two state variables.



9.4 MATHEMATICAL DETAILS

◎ Eigenvectors - Example

$$\mathbf{A} = \begin{pmatrix} 3 & -6 \\ 2 & -5 \end{pmatrix} \quad \text{then} \quad \lambda_i = (-3, 1)$$

To find the eigenvectors for $\lambda_2 = 1$

$$\begin{pmatrix} 3-1 & -6 \\ 2 & -5-1 \end{pmatrix} \begin{pmatrix} c_{21} \\ c_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (9.48)$$

$$2c_{21} - 6c_{22} = 0$$

$$2c_{21} - 6c_{22} = 0$$

$$\text{choosing } c_{21} = 1, \quad c_{22} = \frac{1}{3} \quad \text{or,} \quad \mathbf{c}_2 = (1, 1/3)$$

Another quick method to calculate the eigenvectors

$$\begin{pmatrix} 1 \\ \frac{\lambda_1 - a_{11}}{a_{12}} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ \frac{\lambda_2 - a_{11}}{a_{12}} \end{pmatrix} \quad (9.49)$$

